Analytics Specializations and Applications

BUSI4392 (2025-2026)

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1. Executive Summary

Task Description:

The goal of this analysis is to segment the customer base of a national convenience store chain to provide actionable insights for targeted marketing, product promotion, and loyalty strategies. This involves analysing customer purchase behaviour, demographics, and engagement patterns to identify distinct groups within the customer population.

Technical Approach:

We employed a data-driven clustering approach to group customers based on selected features representing spending behaviour, frequency, and demographic characteristics. Preprocessing included loading the files and checking the integrity of the values (no nulls, formatting, outliers). Feature engineering is done to select the best features where we divide it into three big categories: customer behavioural features, basket features, and categorical features.

In total we are using 9 features. All features are log transformed, scaled to zero mean and unit variance to ensure that no single feature dominates the clustering due to scale differences. PCA was done to keep 80% of component variants, this is done to create clean silhouette and make the clusters fit better. PC 1 (29.4%) & PC 2 (21.2%) explained enough on the data clustering, totalling up to 50.6% variance explained in the 2D chart. Clustering was conducted using K-means. The optimal number of clusters was determined through Elbow method, Silhouette analysis, and Stability testing of Adjusted Rand Index (ARI) while also looking at the interpretability for business.

Data Summary:

The analysis uses a dataset of 3,000 customers, 195,547 baskets dataset, 1,461,315 items dataset, and 3,000 categories dataset. Combining the four csv datasets into one comprehensive dataset with the customer id number as the primary key. Results included a csv file with all of customer id numbers with their respective categorized segments.

Results Summary:

The clustering analysis produced 5 distinct customer segments, each with unique behavioural and demographic profiles. Two segments emerge as the most strategically important due to their strong revenue potential and engagement with the retailer. Segment 3 - Convenience Store Shoppers (19.65%) are our engagement engine with 17% revenue. They interact with us almost daily, buying small treats. Segment 0 - Everyday Treat Shoppers (25.59%) are our profit engine. They buy high-margin indulgence products and gift items totalling to 26.6% of revenue. Our goal is to increase their frequency. Segment 2 - The Core Household Shoppers (47.45%) are our foundation. We protect them with reliable service and competitive prices as they contribute to 48.9% of revenue. The two small lifestyles of segment 1 & 4 (7.3% combined) are our niche category. They contribute around 3% of revenue each. We want segment 1 to set trends.

2. Feature Description

The selected features were designed to capture multiple dimensions of customer behaviour relevant to retail segmentation. These include customer behaviour, basket behaviour, and categories preferences. Together, these features provide a comprehensive behavioural representation of customers, enabling meaningful segmentation without overfitting. The segmentation reflects both spending and lifestyle patterns.

- **Essentials:** BAKERY, DISCOUNT_BAKERY, DAIRY, FRUIT_VEG, MEAT, FROZEN, GROCERY_FOOD
- **Indulgence:** CONFECTIONARY, SOFT_DRINKS, DRINKS, SEASONAL_GIFTING
- **Convenience:** DELI, PREPARED_MEALS PRACTICAL_ITEMS, CASHPOINT
- **Lifestyle:** WORLD_FOODS, GROCERY_HEALTH_PETS, LOTTERY, TOBACCO, NEWSPAPERS_MAGAZINES

Feature	Description & Justification
total_spend	Total amount spent by the customer across all transactions, representing overall economic value to the retailer.
baskets	Number of shopping visits made by the customer, indicating purchase frequency and engagement level.
recency_days	Number of days since the last purchase, indicating current customer activity and potential churn risk.
essentials_ratio	Share of spending on essential goods, identifying routine household shopping behaviour.
indulgence_ratio	Proportion of spending on indulgent or discretionary products, reflecting lifestyle or premium consumption.
convenience_ratio	Share of purchases from convenience categories, indicating quick top-up or time-driven shopping behaviour.
lifestyle_ratio	Proportion of purchases in lifestyle categories, suggesting engagement with premium or discretionary goods.
basket_spend_mean	Average of basket spending, distinguishing premium shoppers or budget shoppers
basket_quantity_mean	Average of basket item quantity, distinguishing big weekly shoppers from small shoppers

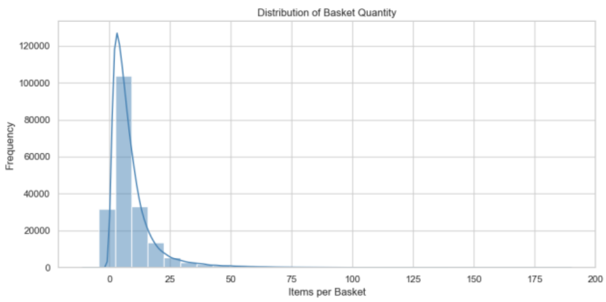
Feature	F Score	P Value
convenience_ratio	931.979716	0.000000e+00
basket_quantity_mean	906.649403	0.000000e+00
lifestyle_ratio	776.907355	0.000000e+00
baskets	737.471036	0.000000e+00
basket_spend_mean	699.931449	0.000000e+00
total_spend	450.317944	3.661299e-304
recency_days	401.064437	6.312838e-277
essentials_ratio	216.496921	2.344766e-163
indulgence_ratio	85.321348	1.021601e-68

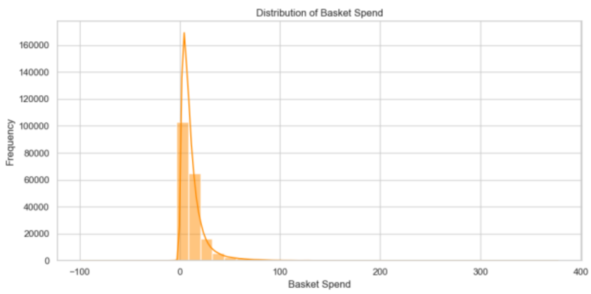
3. Customer Base

The customer base exhibits significant variation in spending, shopping frequency, and purchasing patterns. Total spend ranges from occasional shoppers to highly engaged customers, while the number of store visits and average spend per basket highlight differences between quick small trips and larger planned purchases.

Product preferences also vary. Most spending focuses on essential goods, reflecting the retailer's core role, but a segment prioritizes discretionary or indulgence items, while others prefer convenience-oriented products. Return transactions are removed to focus on purchase behaviour. Temporal and seasonal behaviour is not captured in this analysis.

Customer engagement differs by recency and shopping consistency. Overall, these insights indicate a heterogeneous customer base with distinct behavioural patterns, supporting the use of segmentation for targeted marketing and strategic planning. Basket size and spend distributions are balanced with a little left skew. However, it shows minimal outliers and most baskets are around in the same size and spend. What makes them different will be the preferred item categories.





4. Segmentation Methodology

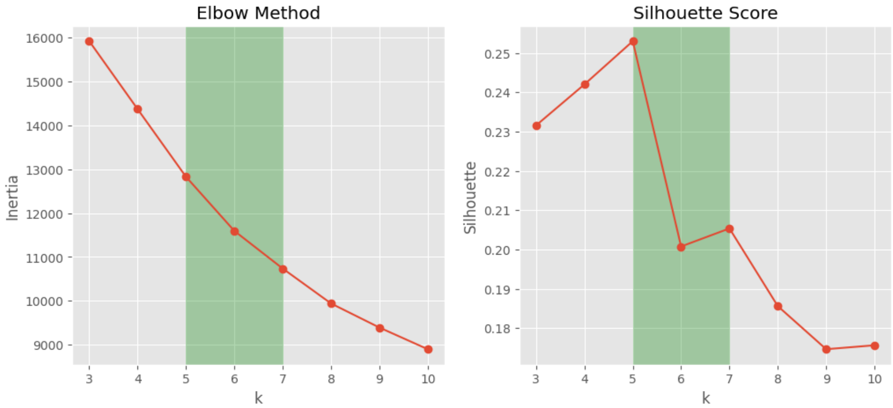
Clustering Approach:

We used K-Means, an unsupervised machine learning method that partitions data into K distinct clusters based on similarity. Which is appropriate for larger datasets, numerical features, and compatibility with PCA. PCA improves cluster separation and reduces noise in the data. Alternative approaches considered included Hierarchical clustering and DBSCAN, but Hierarchical clustering would be too slow and complex to handle thousands of customer data while DBSCAN is less interpretable when the clusters have different densities like these retail customers who usually form gradual behavioural clusters. However, K-Means offered the best balance of interpretability and stability.

Determining Number of Clusters (k):

The optimal number of clusters was selected Elbow method, Silhouette analysis, and Stability testing of Adjusted Rand Index (ARI), where k=5 was chosen as it demonstrated high silhouette score (0.253) while maintaining strong stability (ARI = 0.87). This produced distinct, actionable customer segments with minimal overlap.

K=6 and K= 7 both have lower stability and silhouette scores, furthermore its segmentation has 3 or more niche segments (<5% customer population) which makes it harder to interpret with many overlap segments. So, K=5 is chosen for better business interpretability and application.



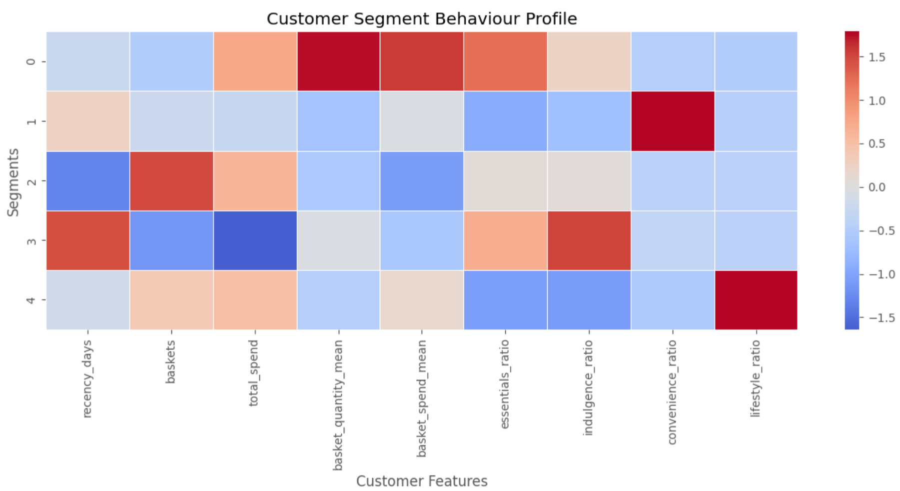
Preprocessing Steps:

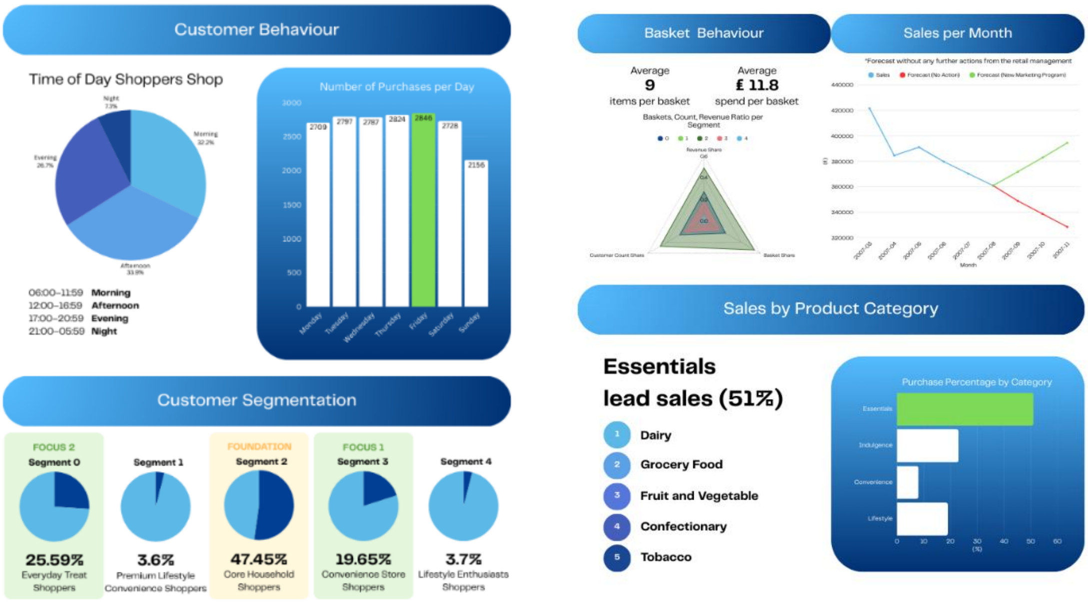
- **Scaling:** All numerical features were standardized to have zero mean and unit variance.
- **Handling Missing Values:** Median imputation but these datasets do not have any null values. 0 imputation for customers who only visited once for gap days. Removing customers who only visited once as they cannot be categorised because they have no patterns.
- **Handling Negative Values:** Negative values is counted as returns or refund. They are removed and put in separate data as it could pose a problem in behaviour analysis segmentation. It could cause double counting error, distribution skew, and feature contamination of total spend or basket quantity. 115 customers have done a refund/return that means 3.8%. It’s tiny and are outliers, not a pattern. The algorithm will get messed up by the noise.
- **Dimensionality Reduction:** PCA was applied to reduce noise and improve clustering quality (80% components). We are left with 5 components. Too many components lead to the dimensionality curse.

5. Analysis & Results

Statistical Insights:

Segmentation separates customers along basket size, spending, product preference, and shopping frequency, enabling tailored marketing, loyalty, and promotion strategies.





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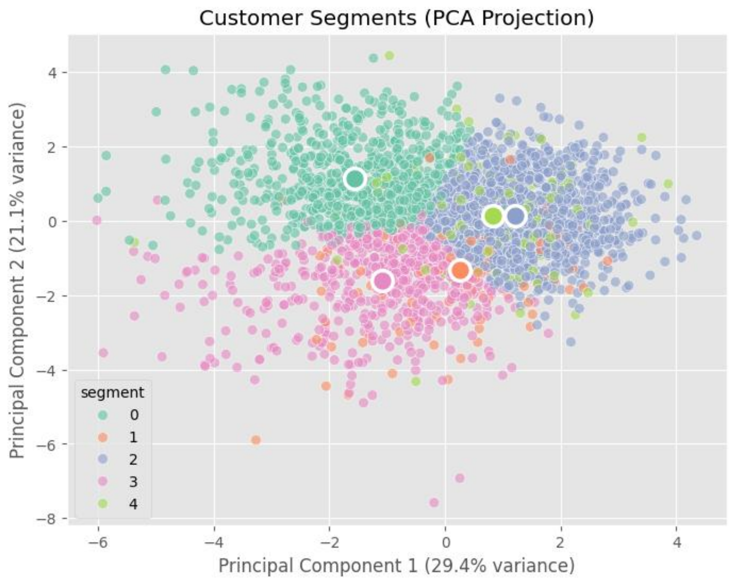
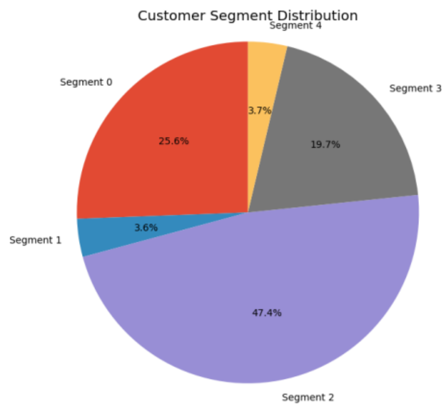
Segment Summary:
Some segments show tight clustering, indicating consistent behaviour. Other segments are more dispersed, reflecting diverse preferences. High-frequency shoppers form clearly distinct clusters. Niche segments appear smaller but highly differentiated.





Segment	Size	Revenue	Key Features	Archetype Name
0	25.59%	26.68%	Moderate shopping with high indulgence products	Everyday Treat Shoppers
1	3.6%	3.48%	High lifestyle items and convenience, has a high basket spend	Premium Lifestyle Convenience Shoppers
2	47.45%	48.97%	Balanced spend, quantity, frequency	Core Household Shoppers

3	19.65%	17.05%	Frequent shopping with low spend and quantity, has a tendency towards indulgence items.	Convenience Store Shoppers
4	3.7%	3.79%	High baskets spend and quantity, very high lifestyle products.	Lifestyle Enthusiasts Shoppers

Descriptive Profiles (Pen Portraits):

- **Segment 0 – Everyday Treat Shoppers:** These customers represent routine shoppers who visit the store regularly to purchase treats and indulgence. Their purchasing behaviour is predictable and stable, with small basket sizes but consistent spending patterns. They rely on the retailer as a convenient source for indulgence. Retailers can engage them through discount bundles and loyalty programs to encourage slightly higher spending with repeat visits.
- **Segment 1 – Premium Lifestyle Convenience Shoppers:** Customers in this segment shop regularly with a larger basket sized and spend amount. Very high lifestyle and convenience products. They are a high value niche segment that has little time to do shopping as suggested in their product interests that consists of ready-to-eat foods and convenience. They will react well to personalized promotions, loyalty programs, and VIP programs.
- **Segment 2 – Core Household Shoppers:** These customers are the core loyal customers. They shop regularly with average basket sized and spend amount. Maintaining a balanced interest in multiple categories Their visits are often planned for weekly, biweekly or monthly purchases. This segment can be targeted with bulk discounts, family pack item promotions, and subscriptions options to encourage them to regularly shop.
- **Segment 3 – Convenience Store Shoppers:** Customers in this segment appear to prioritise speed and ease, making small quick trips to buy indulgence products. Their baskets are low in volume with high percentage of convenience items but very frequent consistent shoppers. They are price sensitive, hence low spend and quantity per basket. To boost this segment, retailers can introduce express checkouts, pre-order and ready to pick-up services, self-checkout counters with convenience item deals to enhance their satisfaction and boosting repeat visits.
- **Segment 4 – Lifestyle Enthusiasts Shoppers:** This segment reflects customers who are driven by emotion and lifestyle preferences. They buy treats, premium products, and gifts with very little number of essentials. However, they have less visits than other segments, making them quite inactive shoppers and at the lower end of engagement. These are niche enthusiasts. They're deeply invested in one or more lifestyle. Seasonal marketing and premium product promotions will be effective in getting their attention to boost spending amount.

6. Summary & Recommendations

Key Results:

The clustering analysis revealed 5 distinct customer segments, each with unique purchasing behaviours and demographic profiles. This segmentation allows company to tailor marketing, loyalty, and product strategies for higher ROI. Pushing for focus on segment 3 and 0.

Recommended Focus Segments:

1. **Segment 3 – Convenience Store Shoppers (19.65%):** Due to their high frequency visits, they are doing emotional attachment to their bought products for personal indulgence. Campaigns targeting this group are likely to provide the most ROI. The company should prioritize loyalty reinforcement strategies:
   Strategic Goal: Increase basket size while maintaining frequency.
   - Bundles of indulgence products
   - Subscriptions of their daily bought items
   - Time-based offers by adding breakfast items or lunch deals, etc
   - Free delivery threshold raised slightly to encourage basket size (Online Exclusive)

2. **Segment 0 – Everyday Treat Shoppers (25.59%):** While not yet fully loyal, these customers are buying high-margin products. Their shopping behaviour can be predicted nearing the seasonal holidays as they spend more on special occasions.
   Strategic Goal: Increase frequency and capture gifting occasions.
   - Occasion-based marketing (Valentines, Birthday, Christmas, etc)
   - New product alerts for limited-edition treats
   - Bundle offers for seasonal items
   - Loyalty program with tiered benefits

Other Segments Insight:

1. **Segment 2 – Core Household Shoppers (47.45%):** Maintain with reliable service and competitive pricing. They are the retail company foundation.
2. **Segment 1 – Premium Lifestyle Shoppers (3.6%):** Niche but has high value, offer then new premium lifestyle products and partnerships to promote them as trendsetters.
3. **Segment 4 – Lifestyle Enthusiasts Shoppers (3.7%):** Ensure that the specific lifestyle products are stocked. They are a small population but are loyal within their niche. Gather information from them on popular lifestyle products are in.

Additional Recommendations:

As a retailer you might want to focus on the top performing segments, but we'd like to recommend you consider engaging with other segments who has low to moderate frequency and niche interests.

- **Marketing:** Tailor promotions and product recommendations according to cluster preferences. Such as, using emotional wording such as *"Treat yourself to…"*.
- **Digital Marketing:** Sync between online and offline purchases, with mobile or email push notifications such as *"You may be running low on…"* and getting digital coupons for purchases made in stores to simultaneously boost mobile app (online) sales. Time-based marketing notifications for early bird, late owl promotions. If possible, include weather-based marketing that can suggests customers to buy weather related products (Soupy foods during rainy days, ice creams or cold drinks during hot sunny days, etc).
- **Loyalty Programs:** Design tiered incentives (collect stickers, coupons) to encourage higher spend and frequency among low to medium engagement clusters. Referral system to bring in more customers can also be considered.
- **Trends:** Jumping on trending activities and items, for example blind box merchandise (key chains, small trinkets) or freebies with purchase conditions (hand fan, tote bag) to attract low to moderate engagement shoppers with events like *"buy specific products to get freebies"*.
- **Seasonal Analysis:** Explore seasonal purchase patterns, response to promotions, holidays and big events purchase tendencies according to local area habits and customs.
- **Extra**: The retailer can introduce a monthly "Essentials + Surprise" box with options of subscriptions. The box would feature a core of essential items (≈70–80% of contents) aligned with their primary purchasing habits, complemented by indulgence and lifestyle products (≈20–30%) as surprise bonuses. This approach converts occasional shoppers into predictable subscribers, enhances perceived value, encourages discovery of new products, and helps manage excess inventory, turning a low-engagement segment into a steady revenue stream.

Conclusion:
By leveraging these insights, the company can optimize resources and business impact. Focus on Segment 3 (Convenience Store Shoppers) to increase basket size and Segment 0 (Everyday Treat Shoppers) to increase frequency. Protect and stabilize Segment 2. Monitor Segments 1 and 4 for future trends.